Worksheet 56 · Histograms IGCSE

Edexcel IGCSE Mathematics · past paper questions

Name:	Subject: Math	Date:
Learning Objectives: Students will be able to: • Calculate frequency density • Draw histograms with unequal class widths • Find frequencies and estimates from a histogram		
Grade: / 46	**Teacher's Feedback:**	

Answer in the space given. Show your working. Marks for each part are shown in brackets.

Key notes

Formula → Frequency Density = $\frac{\text{Frequency}}{\text{Class width}}$

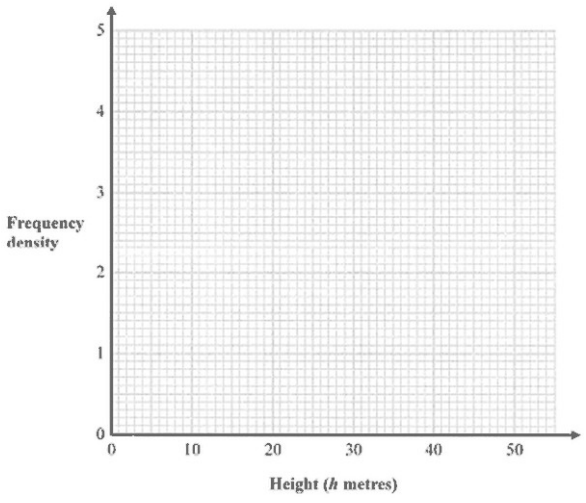
Questions *Histograms*

Question 1 (3 marks)

The table gives information about the heights of some trees.

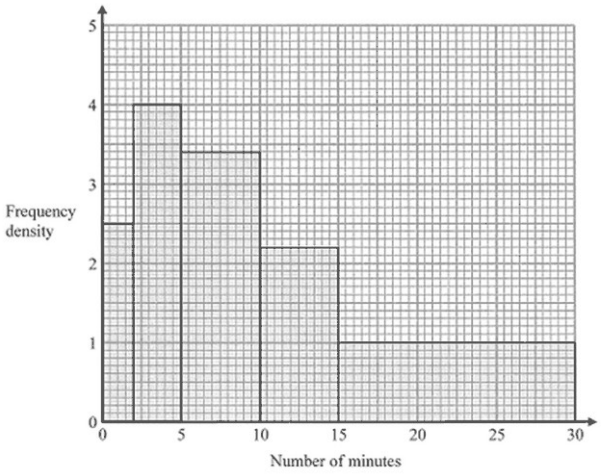
Height (h metres)	Frequency
$0 < h \leqslant 20$	15
$20 < h \leqslant 35$	48
$35 < h \leqslant 40$	21
$40 < h \leqslant 50$	16

On the grid, draw a histogram for this information.



Question 2 (3 marks)

The histogram shows information about the numbers of minutes some people waited to be served at a Post Office.



Work out an estimate for the proportion of these people who waited longer than 20 minutes to be served.

Question 3 (5 marks)

17 The histogram shows information about the ages of all the passengers travelling on a plane.
No one on the plane is older than 80 years.



24 passengers on the plane are aged between 40 years and 60 years.

(a) Work out the total number of passengers on the plane.

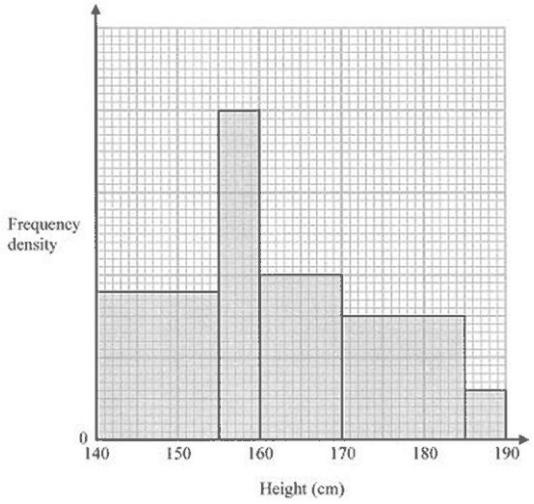
(3)

A passenger on the plane is picked at random.

(b) Work out an estimate for the probability that this person is older than 55 years.

(2)

Question 4 (4 marks)



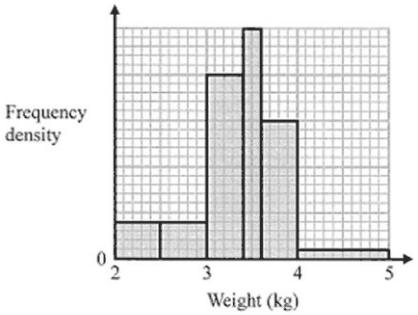
The histogram gives information about the heights of all the Year 11 students at a school.

There are 160 students in Year 11 with a height between 155 cm and 170 cm.

Work out the total number of students in Year 11 at the school.

Question 5 (3 marks)

The histogram shows information about the birth weights of some babies.



6 of these babies had a birth weight less than 2.5 kg or greater than 4 kg.

Work out the number of babies who had a birth weight between 2.5 kg and 4 kg.

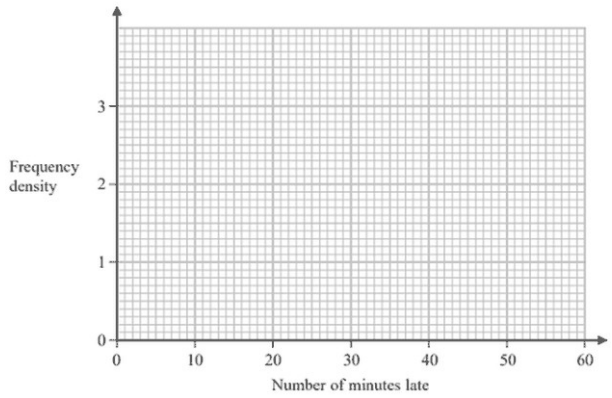
Question 6 (5 marks)

Manuel collected information about the flights that arrived late at an airport last month.

The table gives information about the number of minutes that these flights were late.

Minutes late (L minutes)	Frequency
$0 < L \leqslant 10$	8
$10 < L \leqslant 15$	13
$15 < L \leqslant 25$	19
$25 < L \leqslant 40$	24
$40 < L \leqslant 60$	6

(a) On the grid, draw a histogram for this information.



(3)

Manuel selected at random a flight that was late by 25 minutes or less from his results.

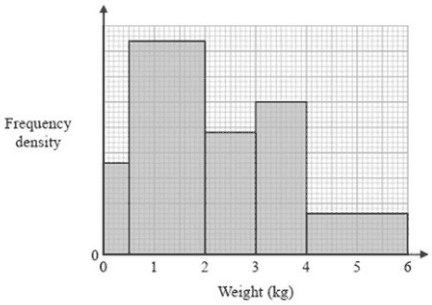
(b) Work out an estimate for the probability that this flight was late by 5 minutes or less.

(2)

Question 7 (6 marks)

A postman records the weight of each parcel that he delivers.

The histogram shows information about the weights of all the parcels that the postman delivered last Monday. No parcels weighed more than 6 kg.



63 of the parcels that the postman delivered last Monday each had a weight between 0.5 kg and 2 kg.

(a) Work out the total number of parcels the postman delivered last Monday.

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(3)

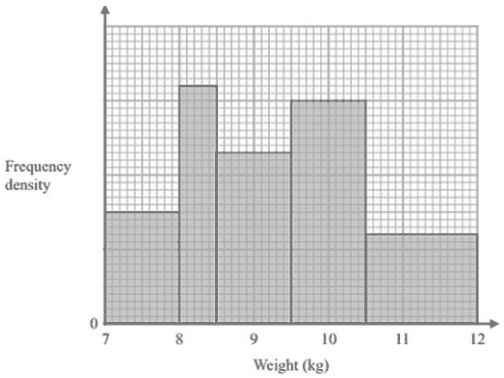
The postman picks at random two of the records of the parcels he delivered last Monday.

(b) Work out an estimate for the probability that each parcel weighed more than 2.25 kg.

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(3)

Question 8 (3 marks)



The histogram gives information about the weights, in kg, of all the watermelons in a field.

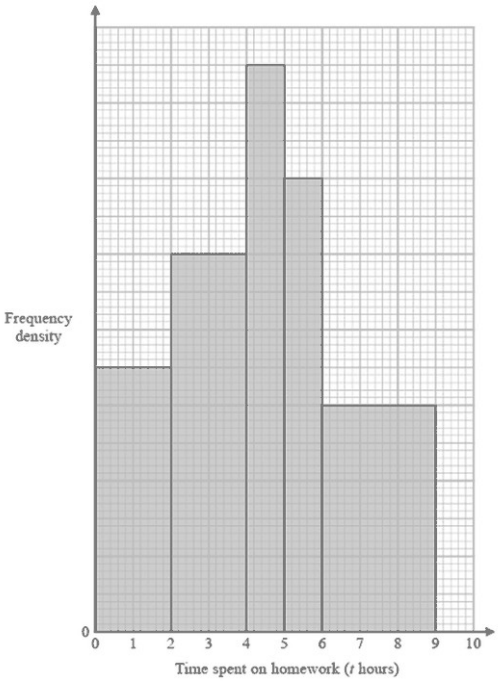
There are 16 watermelons with a weight between 8 kg and 8.5 kg

Work out the total number of watermelons in the field.

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Question 9 (5 marks)

The histogram and the table give some information about the amounts of time, in hours, that Year 11 students at Bergdesh Academy spent, in total, on their homework last week.
No student in Year 11 spent longer than 9 hours on their homework.



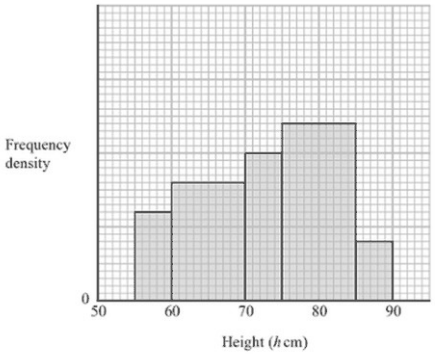
Time spent on homework (t hours)	Frequency
$0 < t \leqslant 2$	28
$2 < t \leqslant 4$	
$4 < t \leqslant 5$	
$5 < t \leqslant 6$	
$6 < t \leqslant 9$	

Using the information in the histogram and in the table, work out an estimate for the mean amount of time the Year 11 students spent on their homework last week.
Give your answer in hours correct to 3 significant figures.

..... hours

Question 10 (4 marks)

The histogram gives information about the heights, h cm, of some tomato plants.



There are 12 tomato plants for which $75 < h \leqslant 85$
One of the tomato plants is selected at random.

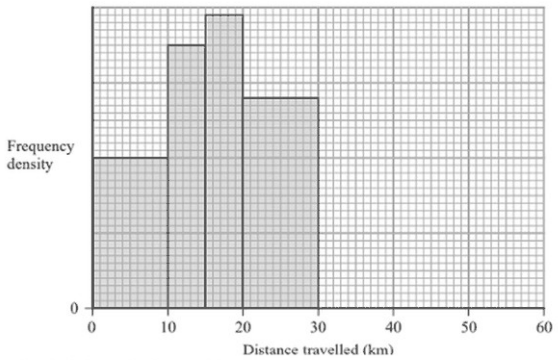
Find an estimate for the probability that this tomato plant has a height greater than 82.5 cm

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Question 11 (5 marks)

The table and histogram give information about the distance travelled, in order to get to work, by each person working in a large store.

Distance (d km)	Frequency
$0 \leqslant d < 10$	40
$10 \leqslant d < 15$	
$15 \leqslant d < 20$	
$20 \leqslant d < 30$	
$30 \leqslant d < 60$	30



Using the information in the table and in the histogram,

(a) complete the table,

(2)

(b) complete the histogram.

(1)

One of the people working in the store is chosen at random.

(c) Work out an estimate for the probability that the distance travelled by this person, in order to get to work, was greater than 25 km.

(2)